

Engineering and Scientific Activities/Studies in Antarctica

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Abstract

An emergent requirement for a new power generation and supply system had to be developed and installed in Antarctica during XII Indian Expedition to meet the demand of Indian Research Station 'Maitri' since the existing facilities have been damaged due to fire accident. Considering the Antarctic conditions, easy construction and transportability by helicopter, three shelters of knock down type containers made out of prefabricated components and panels were designed and installed. The demand for the power house was received in R&DE (Engrs), Pune on 16 Sept 92 and the power house 'Aditya' was commissioned in Antarctica on 18th Feb 93 in a record time. The main factor that affect habitat construction in Antarctica are low temperature, high wind speeds, snow accumulation and snow drift, regional surface characteristics, visibility and accessibility/inaccessibility. During wintering period, studies on behaviour of material used for construction of buildings, effects of Antarctic environment on foundation and structures, effectiveness of heating and ventilation system, DRDO experimental HF Data Communication system, power and water supply system were carried out.

Introduction

The Research & Development Establishment (Engineers), Pune is one of the establishments under DRDO. The association of R & DE (Engrs), Pune with the Indian Antarctic Expedition started from II expedition when indigenously manufactured structures/equipments were sent to Dakshin Gangotri (DG) to try in the winter period. Based on the feed back and requirements projected by wintering teams, improvements/modifications were carried out and additional structures and equipment were sent in the following expeditions.

In continuation of the support, R&DE (Engrs) have designed and constructed Maitri, India's second permanent station during VII and VIII Expedition and a Power House in X Expedition with fully indigenous material and equipments. The generating sets were commissioned in XI Expedition but due to fire accident in Power House, a requirement for a new gen sets accommodation and gen sets were found essential to overcome the critical situation.

Besides the various tasks for XII Expedition and X Wintering Team, the major task was to construct new accommodation for gen sets and commissioning of the same at the earliest.

Representative of R&DE (Engrs) was assigned the task to provide technical assistance and assist the Army Team in construction of the new accommodation, installation and commissioning of 4×62.5 kVA gen sets in addition to carrying out maintenance of life supporting equipments/services at Maitri. He was also entrusted with carrying out Scientific Studies during wintering period on behaviour of engineering material used for the construction of buildings, effects of Antarctic environment on foundation and structures, improvement in green house design, heating and ventilation system, HF voice and data communication system, power supply and sewage disposal system.

Aditya Power House

Maitri station when commissioned in 1988-89 had been provided with 4×62.5 kVA gen sets for power supply. Since these gen sets have been outlived their lives and to avoid noise and other pollutions to main building, an elaborate power module with adequate space and volume was constructed and commissioned 4×62.5 kVA and 2×125 kVA gen sets during XI Expedition. However, due to fire accident the facilities have been damaged. Hence the new requirement of power supply system on emergent basis had to be developed and installed at Maitri.

Requirement from the users regarding the new power supply system was received in R&DE (Engrs), Pune on 15 Sept 92 and the complete system was to be designed, manufactured and despatched to Goa for shipment before first week of Dec 92. This could be done with the vast experience gained during the participation of previous expeditions and untiring efforts by a team of officers and staff of R&DE (Engrs), Pune. Considering the emergent demand, the task was taken up on war footing and the power house Aditya could be commissioned on 18th February, 1993 in Antarctica.

Gen Sets Accommodation

Gen sets accommodation was designed in such a manner that it can cater for housing 2×62.5 kVA gen sets in each shelter. Three types of knock down type containers named Type A, B and C (Figs 1, 2, and 3) having overall size of $9 \text{ m} \times 3 \text{ m} \times 3 \text{ m}$ (approx.) similar to ISO containers were provided, two for housing 2×62.5 kVA gen sets in each shelter and one as control room-cum-workshop. These three containers were to be joined in Y shape by a central

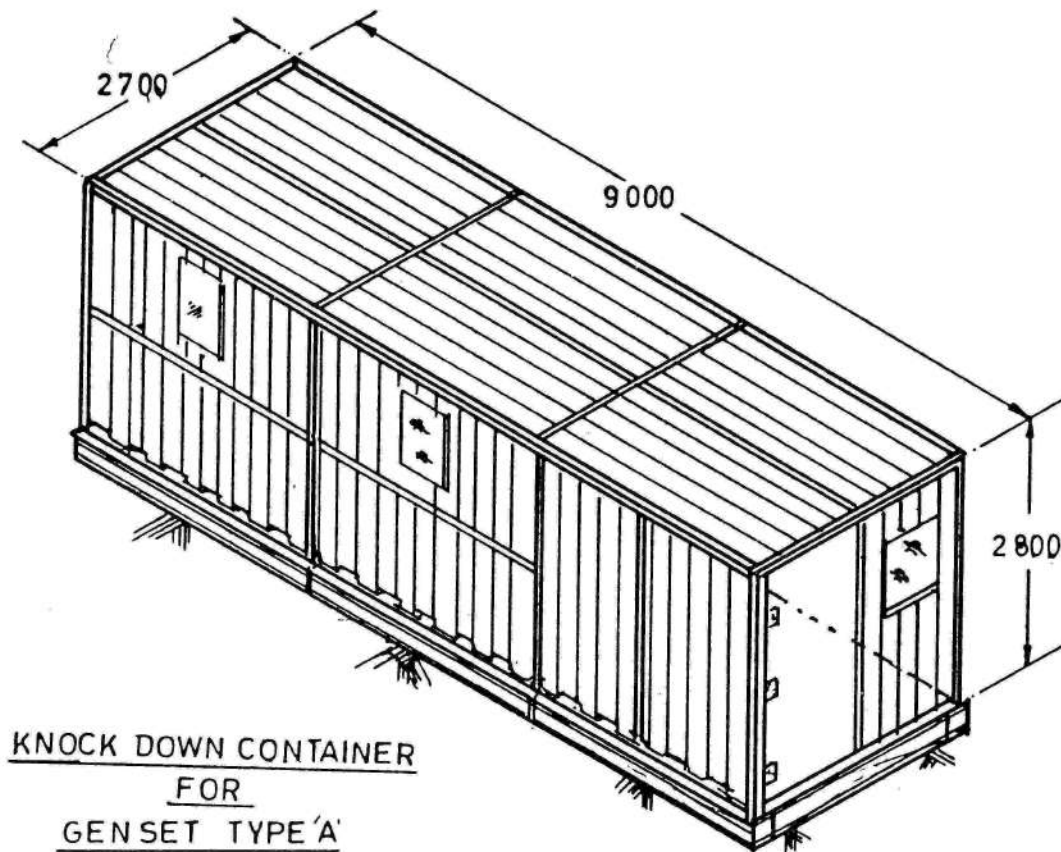


Fig. 1

hexagonal connector module as shown in Fig. 4. Keeping in view of the size and weight restrictions for carriage by MI-8 helicopter and easy construction at site, the knock down containers were made out of prefabricated components and panels when assembled will form the accommodation. The maximum weight of an individual panel/component was restricted to 100 kg. Each container was split into small structural elements and interconnected by nuts/bolts and special types of sockets at each junction. These containers were designed to construct on grillage beam foundation anchored to concrete blocks. Resin bonded fibreglass packed in polythene bags as per the panel size had been provided for insulation of each type of container. These bags were placed at their proper positions and closed with inner lining of fire retardant sheet, Bison Panel, which were cut into sizes and screwed into position. Filling of all the gaps from inside and outside of each container were done by means of silicon sealant provided in tubes operated by sealant gun.

The various tasks involved were:

- (a) Selection of site
- (b) Levelling/preparation of ground

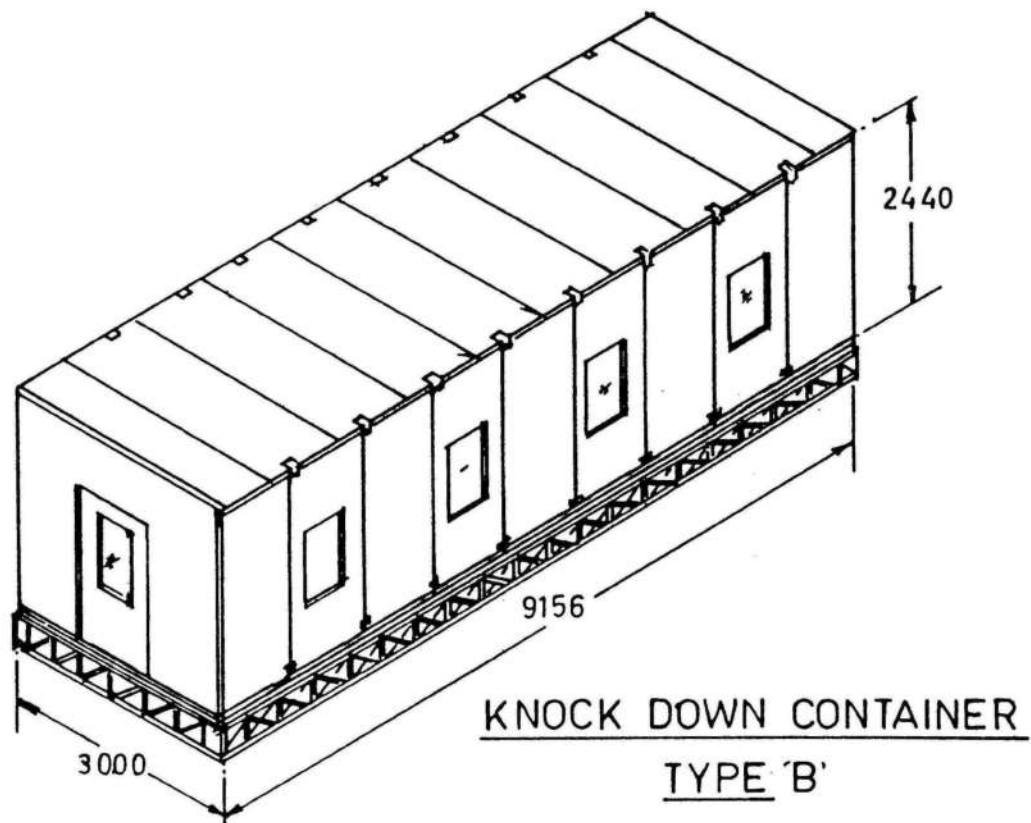


Fig. 2

- (c) Foundation work
- (d) Assembly of individual beams
- (e) Assembly of basic structure
- (f) Joining the containers with central connector module
- (g) Insulation and fire proofing
- (h) Installation of accessories
- (i) Internal wiring/electrical fitments
- (j) Installation of fire protection sensors/extinguishers

IX Wintering Team selected a site for the construction of accommodation approx. 400 m away from the main station. Considering the various aspects like providing fuel line, main supply cable to distribution board, voltage drop, routine inspection, attendance, maintenance etc., a new site was selected approx. 100 m away from the existing main distribution board and its ground was levelled (Fig. 5). Utmost care has been taken in levelling the ground and accurate foundation work which helped in easy and trouble free assembly of super structure of knock down containers and hexagonal connector module.

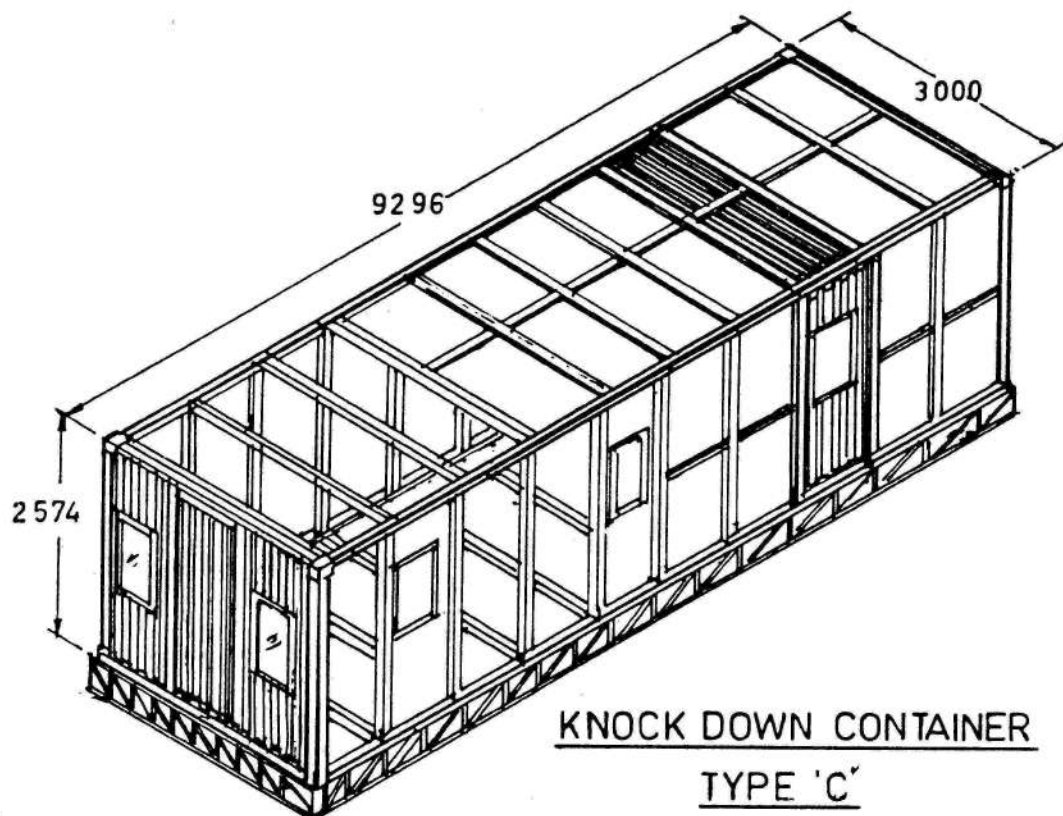


Fig. 3

Complete construction work had been carried out as per the design and laid down procedure of R&DE (Engrs).

Elaborate planning for construction was carried out during the voyage and construction activity started on the first day, *i.e.*, 1st Jan 93, of landing at Maitri by the new team. When the temperature falls down in winter, the upper layer of the earth becomes very hard due to frozen soil and digging of any sort is difficult. Such work can be carried out only in summer period. Digging of pits for foundation was started but the weather became bad and heavy snowfall experienced for the next four days, as such the construction material could not be transported from ship to Maitri by helicopter. The pits were completely covered with snow and had to be removed frequently before concreting the foundation. The construction material started arriving w.e.f. 8 January, 1993. On good weather day, the construction team worked more than 12 hours a day to complete the task within the specified time. The central connecting module had to be modified and an additional space was to be constructed to accommodate the kerosine fired heating system provided for the shelters.

Out of the three different container modules, the modular panel design, type C, was found to be easy in construction considering the Antarctic condi-

'ADITYA' POWER HOUSE

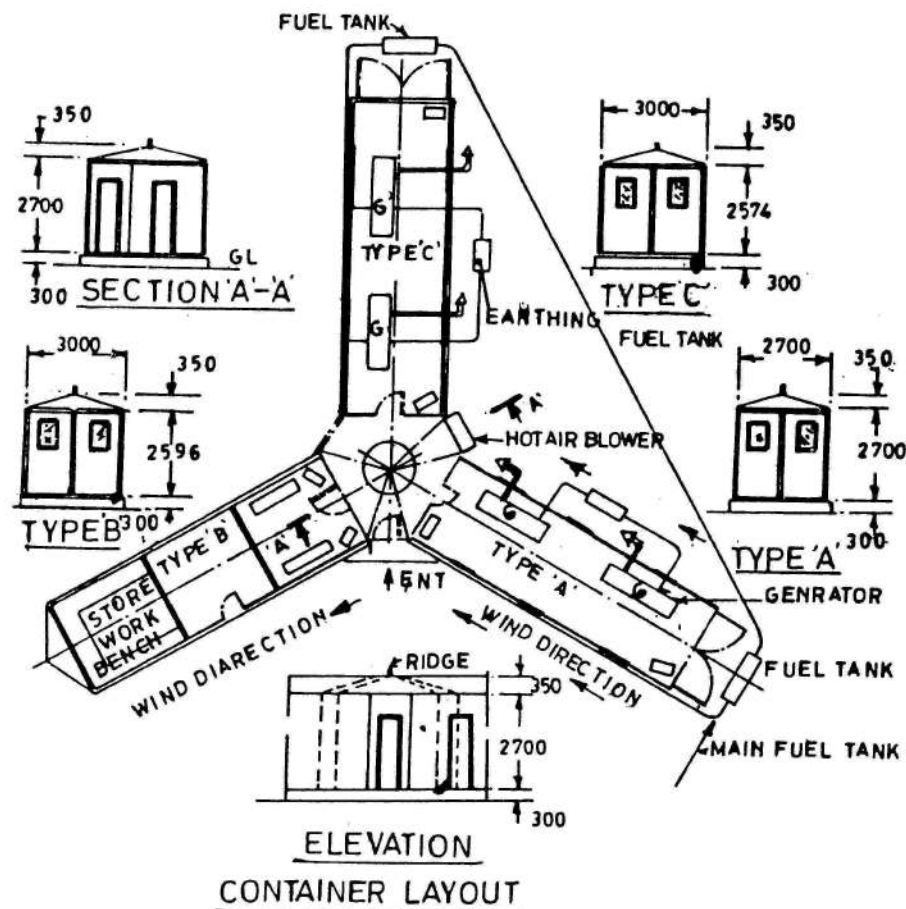


Fig. 4

tions and available manpower. Comparing with other two containers, there were no columns, edge runners, cross runners and less hardware such as nuts, bolts etc. were used in assembly and construction. This was also found to be more leakproof, easy to provide sealing and less prone to snow ingress.

Hydraulic binder, special type of cement, was tried for concreting the foundation pits for anchoring grillage beams of one container and found quite useful in Antarctic conditions.

For inside finishing of walls, roof and to avoid fire hazards, fire retardant Bison panels were properly fixed. Cavity and gaps were filled with silicon sealant.

Proper internal wiring of the containers was carried out with correct size of PTFE Teflon coated cables with casing and capping and adequate lighting was provided.

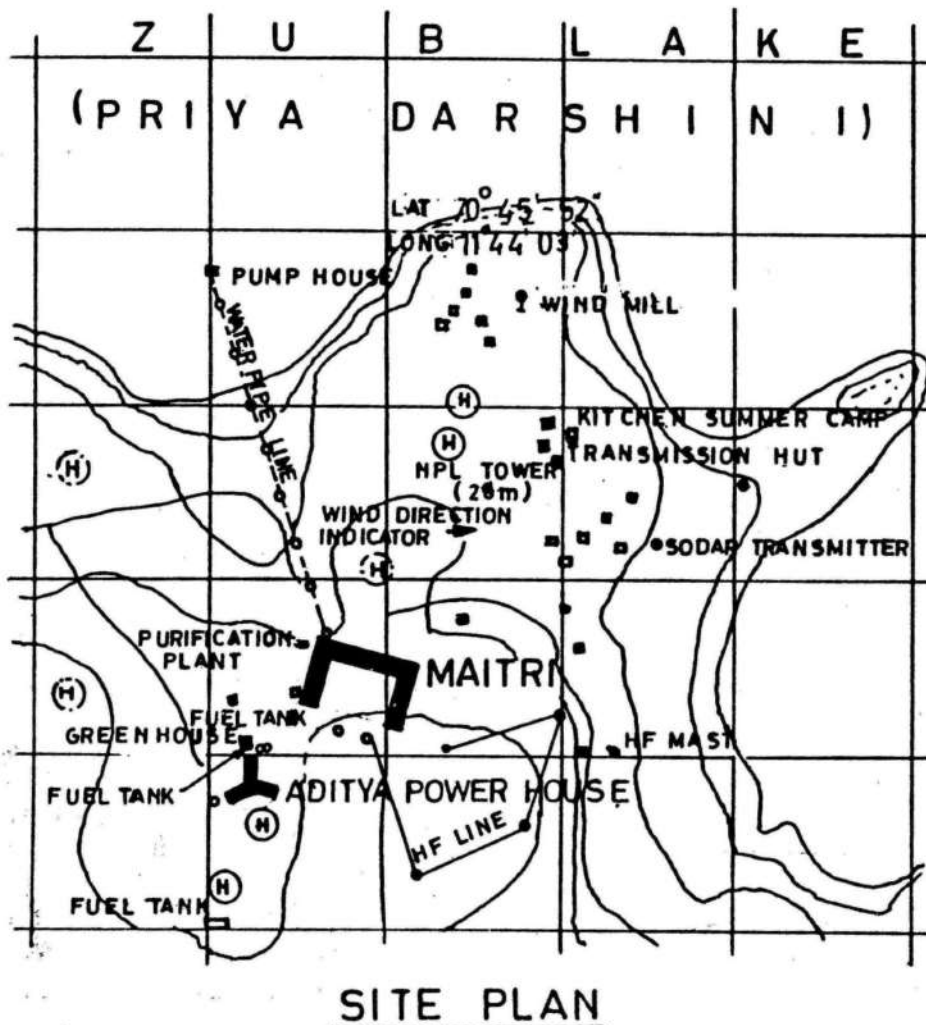


Fig. 5

Inside the containers, suitable fire protective devices have been installed at appropriate places.

Installation and Commissioning of 4×62.5 kVA Gen Sets

The task involved the following :-

- (a) Fixing gen sets mounting channels with DUNLOP cushy foot antivibration mountings
- (b) Placing gen sets and bolting to the mountings
- (c) Provision of exhaust system
- (d) Provision of fuel system

- (e) Wiring and fitment of cables with proper plugs, connectors and inter-connection of control panels
- (f) Provision of proper earthing
- (g) Proper testing and commissioning of gen sets.

As and when the base frame and the flooring of the containers were ready, the gen sets mounting channels were fixed. To isolate the vibration while running of the gen sets, cushy foot antivibration mountings were fitted to the mounting frame. Generating sets from the ship were brought only when the flooring of accommodation was ready to avoid them from exposure to lower temperature conditions of Antarctica. The gen sets were placed on the mounting frame and within a day walls and roof of the containers were constructed. The gen sets were then bolted to the foundation frame.

All generators have separate exhaust system. The exhaust pipes were fitted with silencer and outlets were taken outside the accommodation. All the joints were fitted with asbestos lining to avoid leakage of exhaust gas. For ventilation, exhaust fans (wall mounting type) were fitted.

A daily fuel tank of 750 litres capacity has been provided separately to cater for two gen sets. Fuel tank mounting frames were not supplied by us as such were designed and fabricated with the material available in the station. To avoid pumping of fuel to fill the tanks, height of the mounting frame has been designed in such a way that the tanks can be filled by gravity from the main tank otherwise an additional pumping set could have been required for this purpose. The gen sets were provided with fuel lines of copper pipes with necessary control valves. Fuel lines were thoroughly tested before giving connection to the overhead tank.

Each gen set was provided with a control panel near the gen set. Also a remote control panel for each gen set had been provided in control cabin alongwith separate power distribution board installed in Control Room (Type B container). Proper wiring/connections with correct size of cables were made between control panels, remote control panels and distribution board in the control cabin. From the distribution board to the main distribution panel of A block of Maitri station, cables were laid down and secured properly to avoid damage during strong winds and blizzards.

All the gen sets have been provided with adequate earthing as designed by R&DE (Engrs).

After thorough checking as per the instruction of the manufacturers, the gen sets were put on initial running and testing on 14 to 18 February 1993 and the performance were found to be satisfactory. The gen sets accommodation was named as Aditya and inaugurated and commissioned by Dr. V.K. Dhar-

galkar, Station Commander on 18 February 1993 at 1030 hrs. The load of Maitri station was switched over to the new gen sets and subsequently performance were observed throughout the period (Plates 1-4).

It is worth mentioning that the power house could be constructed and commissioned in a record time due to the untiring effort of Army team and support provided by other wintering members. Even though the weather was bad and hostile during the construction period, the work progressed without any hinderance, keeping the emergent requirement of power house in mind by the team. Average minimum temperature and average wind speed varied between -2°C to -7°C and 11.3 to 17 knots with maximum average wind of 42 to 44 knots, during the construction period. Heavy snowfall also prevailed for 10 days.

Scientific Studies

Scientific studies in the field of engineering were carried out during the wintering period.

R&D Programme

Materials

Materials such as steel, mild steel, timber, plywood etc. exposed to low temperature become too hard, and were found to be very difficult to cut. Welding is also difficult. Pipe lines, copper and galvanised iron break open due to thermal expansion and contraction, in case flexible hoses are not provided at the joints.

Effectiveness of the glass wool as an insulation medium was studied by recording the temperature at various places, inside and outside the containers, during the extreme winter period and blizzards. The difference in temperature recorded was 5°C between inside and outside the container.

Hydraulic Binder

Trials were carried out on hydraulic binder, special type of quick setting cement, for concreting the foundation of one of the containers for gen sets accommodation. It was found to be suitable for concrete work at lower temperature conditions keeping the binder slurry temperature between $+12^{\circ}\text{C}$ and $+18^{\circ}\text{C}$. The concrete block was completely set within 24 hours. When the ambient temperature is low, maintaining the water and slurry temperature and mixing in open space becomes difficult.

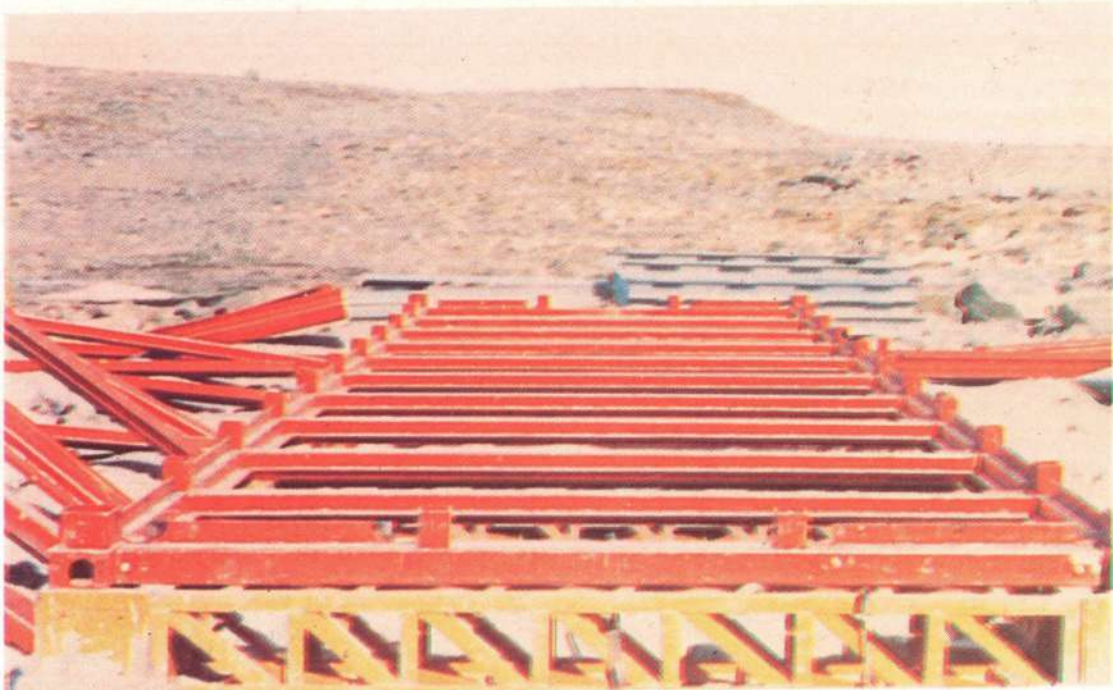
Plate -1

The new Power House 'ADITYA' at Maitri

Plate -2



Construction material transported by MI-8 Helicopter



Grillage foundation work in progress

Plate -3



Assembly of knock-down container (Control Room)



'ADITYA' Control Panels

Plate -4



*Inauguration/commissioning of 'ADITYA' Power House by Dr. V.K. Dhargalkar,
Station Commander*

Field Test on Composite Material

Specimen of the following were exposed to Antarctic conditions to evaluate the weathering effect for one year period. These samples were taken back to India for further evaluation to find out their suitability for fabrication of structural components.

- (i) Two hand trays made out of glass fibre in two resins, namely Exposy and Polyster.
- (ii) Three ropes of different materials such as nylon, polypropylene and braided glass rovings put on a frame to keep the ropes under tension.

It has been observed that there was no considerable change/effect due to the exposure to Antarctic weather conditions.

Static Electric Charge and Earthing

Static electric discharge was observed in Maitri and it was found very high during the winter period. This affects the performance and can cause heavy damage to electronic equipments and components. To protect the equipment against damage, the static electric discharge is to be eliminated.

Static electric discharge control equipments and accessories such as Static area eliminator, Antistatic work kits (table mat, floor mat, grounding hardware, operator's wrist strap) etc. available in the country are to be tried to find out the performance in Antarctic conditions to eliminate the static electric discharge.

In addition to above, it is more important to keep the entire system at the same electrical potential to protect the equipments against damage from static charge. Maitri is built on a rocky area with steel foundation and the steel structure is grounded to the earth. A bare copper conductor of approx. 70 mm² shall be bonded to the steel work around the entire perimeter of the base to obtain low impedance between the two furthest points of the base. The perimeter conductor shall be connected to all metallic components in the base. All exterior cables and pipelines shall be connected to this perimeter conductor at the point where they enter the building. Impedance to earth during summer and winter shall be recorded to study the effectiveness and to modify the system, if required.

Periodic recording of static charge shall be carried out with the help of static charge meter to study the variation during summer and winter period.

Earthing

Main station is properly earthed during initial commissioning of the equipments. However, from safety point of view, the communication equip-

ments/transmitters may be provided with separate grounding by conventional method. Separate earthing shall also be provided for Scientist hut, HF hut, Satcom hut in summer camp area.

To provide earthing for the equipments in snow bound areas, where normal earthing is not possible, an artificial earthing with a box of approx. 1 m × 1 m × 1 m size filled with charcoal, common salt, soil and a copper plate shall be provided and tried out as an experiment to find out the effectiveness.

Behaviour of Structures and Support Facilities in the Antarctic Environment

Structures

Studies on effect of Antarctic environment on existing structure were carried out. The structures of Maitri, Aditya and the 28 m Mast Tower were found sturdy and no deformation was noticed during and after strong high winds, low temperature and blizzards. Meteorological data recorded during the wintering period is given in Table 1. Due to weathering effects, wooden frame work developed gaps at certain places and snow ingression take place during the blizzards. This was noticed mainly in toilet block, green house, corridor in front of boiler room and roof side of A block. This affects the temperature condition inside the Maitri station.

Table 1: MET Data

Month	Temp (deg. C)			Wind Knots			Blizzards	Snowfall days
	Max.	Min.	Avg.	Avg.	Max. Gust	Above 93 (No.of days)		
All 1993								
Jan	8.2	-5.5	0.7	11.3	48	4	0	7
Feb	5.4	-13.8	-3.5	17.0	57	13	0	8
Mar	-0.1	-19.7	-7.5	21.0	63	25	0	4
Apr	-0.3	-24.5	-12.6	17.5	62	21	1	3
May	-2.3	-26.6	-13.3	18.0	58	23	3	9
Jun	-3.3	-30.0	-15.2	18.3	70	17	2	8
Jul	-7.1	-33.5	-19.8	15.6	52	17	6	13
Aug	-5.0	-34.5	-16.9	19.7	92	23	6	9
Sep	-6.1	-24.8	-14.3	22.0	80	26	4	12
Oct	-2.8	-21.0	-10.4	22.0	86	26	5	13
Nov	3.8	-15.8	-5.5	18.4	72	20	1	3
Dec	5.6	-8.7	-0.4	18.2	67	15	0	11

During winter, some difficulty was experienced in using the toilets due to snow ingress and low temperature inside the toilet block. Structure needs strengthening and heating arrangements. Exhaust fan which can withstand low temperature of -40°C shall be provided to replace the existing ones.

A block and toilet blocks were constructed on the ground level. During blizzards, snow accumulates upto the roof level on the rear side of these blocks. Engine exhaust outlets gets buried under the snow and needs clearance before starting the gen sets. Supply cable from Aditya to A block also partially gets buried under the snow. Snow accumulation was also noticed in the Klargester area.

Inside the green house, snow ingress takes place through number of gaps developed in the wooden frame work and also from the flooring, as such temperature cannot be controlled inspite of using hot air blower and hot water radiators. At present, plants cannot be grown in winter. Wooden structure needs modification/replacement.

Wooden flooring in the corridor makes lot of noise since the plywood sheet used for flooring started bending due to prolonged use. This causes inconvenience to the occupants and need replacement with suitable material.

Aditya power house is constructed on grillage beam foundation. In spite of effort to seal the gaps, snow ingress takes place inside and snow is to be cleared before starting and running the gen sets. Snow accumulates on the rear side upto the roof level covering the engine exhaust outlets and also on the roof top. When the gen set is running, the snow starts melting and water starts dripping inside the accommodation. Temperature inside the control room falls down drastically and separate heating system is required during the winter. Opening for exhaust fan and air intake were to be completely closed and sealed in winter which causes insufficient ventilation. Since two gen sets with control panels are installed in one container, working space is very less to carry out routine checking and maintenance. The structures constructed on stilts or raised platform shall permit easy passage of snow drift and can restrict snow accumulation.

Heating and Ventilation System

The station building warmed to provide comfortable living and working conditions by a central heating system which consists of four 2 lakh kcal/hr hot water generator connected to 64 numbers of radiators of 1000 kcal/hr capacity by circulating hot water at about 65 to 75°C . Out of these only one is used at a time, the second to meet the peak load, the third standby and the fourth for maintenance.

Considering the room size of living accommodation, the radiator size is quite bulky. Radiator with SS or copper tubing or flat type is suggested. Hot water tanks and expansion tanks in Boiler Room are made of mild steel and due to prolonged use, interior of the tanks are rusted and corroded. These require replacement with SS tanks.

Living accommodation is located in both sides of the main corridor. Both rooms in the rear side are directly facing against the wind direction and are always cooler than the other rooms. In winter almost 5°C difference was observed. Cold wind also enters through the gaps of the flooring in the main block which adds to this. Additional heating system is required for rear side rooms.

Humidity is almost zero and for comfortable living either room humidifiers or a central humidifier is felt essential. Forced air ventilation for removal of odours and carbon dioxide is required since once the door and windows of the station are closed, especially in winter, there is no proper air circulation.

Water Supply System

At present water is being pumped from lake about 255 m away from the main station. The pump house is erected inside the lake approx. 80 m away from the bank. The lake water starts freezing from the end of February. Freezing of lake water was monitored from Mar 93 to Nov 93 and is given in Table 2. The maximum thickness of layer of ice of 1.91 m was observed in the month of Sep. 1993. From the month of October onwards, melting starts and in November thickness was reduced to 1.53 m. The melted ice from the nearby

Table 2: Pattern of Freezing of Priyadarshini Lake

Date	Atmos Temp.	Ice thickness
17.3.93	-16.5°C	38.5 cm
14.4.93	-19.6°C	63.5 cm
29.5.93	-09.5°C	110.0 cm
29.6.93	-21.6°C	132.0 cm
27.7.93	-15.3°C	153.0 cm
19.8.93	-19.7°C	169.0 cm
25.9.93	-14.1°C	191.0 cm
19.10.93	-9.8°C	173.0 cm
09.11.93	+4.2°C	153.0 cm

areas form water streams and flow to the lake. This increases the water level and on 15 Jan 1994, an increase of 520 mm was noticed. The cycle of freezing and melting of lake water and low temperature did not affect the water supply system. Duct heating system also functioned very well.

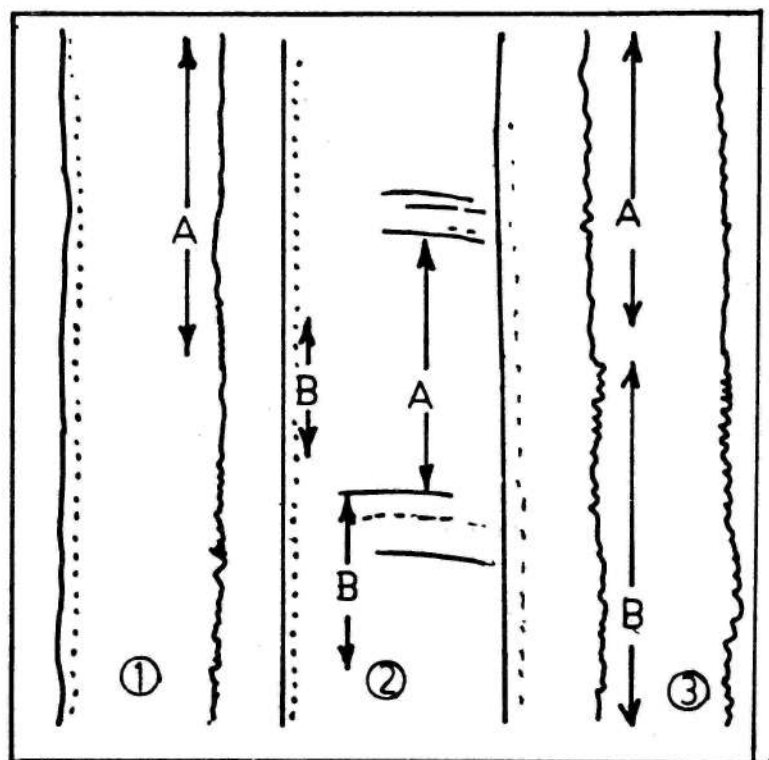
Power Supply

At present 4×62.5 KVA gen sets are available each in Aditya and A block. The gen sets in A block have undergone a number of major repairs and outlived their lives. Two sets are run at a time continuously for 96 hours to cater the power requirement of the station. One genset is connected to boiler room and water supply system and other to the various station loads. To maintain and run these sets at two places is extremely difficult with the available manpower. This also involves more number of cable connections, control panels, change over switches etc. which leads to loose connections and sparking. Higher capacity gen sets with fuel efficient engines can solve this problem.

Naval HF communication is by Morse Code and the duration of transmission is 5 to 6 hours per day. During the transmission, power is drawn at the rate of 10 to 15 kW and the load fluctuation is 2 to 3 times per second. Load fluctuations recorded on 30.9.93 & 25.10.93 are given in Table 3. This sudden fluctuation causes adverse effects on the performance of the gen sets and also all electrical/electronic instruments and sensitive recorders. To avoid malfunctioning and to safeguard the computers, a separate supply had been given from boiler room where the fluctuation is negligible. An example of such recording is given in Fig. 6. This can be avoided in case a separate gen set is run to give supply during the period of Naval HF communication.

Table 3: Load Fluctuation during Naval HF Operation

Date	Time	Power (kW)	Freq (Hz)	Current			Voltage		
				R	Y	B	RY	YB	RB
30.9.93	1100 hrs								
	Normal	25	50.1	40.2	34.8	43.1	415	416	412
	HF Working	40	49.4	53.3	47.5	54.9	417	419	413
25.10.93	1130 hrs								
	Normal	26	50.2	41.5	34.3	43.7	416	417	415
	HF Working	42	50.0	55.7	48.7	59.6	418	420	417



A - RECORDING NORMAL

B - RECORDING DURING HF TRANSMISSION

1 - SURFACE OZONE RECORD

2. TEMPERATURE RECORD

3 - MAGNETIC MICRO PULSATION
RECORD

Fig. 6

Environmental Engineering

Toilet waste from incinerator toilet, burned kitchen waste in the form of ash, metal cans/bottles duly crushed are stored and carried back onboard ship for disposal.

Existing pond for collecting effluent from Klargester was found to be at a lower level and likely chance of overflow to the lake was noticed. A new pond, larger and deeper, has been made with higher side walls to collect the treated waste water which is periodically pumped away to a distance of about 500 m from the pond. Trace heated pipeline has been extended to this pond from Klargester outlet to avoid freezing of waste water inside the pipeline and blockage.

Additional Responsibilities

An experimental HF voice and data communication link between Antarctica and India has been installed during XI expedition (1991-92). Main objective of this experiment is to establish a computer based data communication using HF as the medium. To carry out the experiment during the winter, a representative from DEAL, Dehra Dun was included in the X wintering team but due to medical problems, the name of the individual was dropped. However, Shri Hemant Gaur of DEAL came as a member of XII Summer team and monitored the system during summer. To continue with the experiment during winter, the responsibility was given to the representative of R&DE (Engrs).

DRDO HF communication system consists of transreceiver of 100 watts with a linear amplifier to get a peak envelop power of 1000 watts, a log periodic antenna mounted on a 6 m mast and a radio modem to link the HF equipment with the computer. Different frequencies between 14 to 28 MHz were monitored.

Data and voice communication including phone patch were tried out successfully with R&DE (Engrs), Pune and DEAL, Dehra Dun.

It has been observed that during winter,

- (i) the time slot available for communication is very less.
- (ii) Frequent magnetic storms, local disturbances/ noise and static electricity disturbs the communication.
- (iii) Wind does not affect communication. On trial at a wind speed of 72 knots, clear communication could be established.

The communication equipment is installed in a hut approx. 200 m away from main station. To maintain the inside temperature and to approach the hut during blizzards has been found to be very difficult.

Normal period of Naval HF communication is from 1000 hrs to 1500 hrs. UTC daily and during the period, this system cannot be operated. After 1500 hrs the signal becomes weak and proper communication is not possible.

Further experiment using different type of antennas with increased height, equipments and recorders shall be carried out to ascertain the optimum utilisation.

This system can provide extensive use for the scientists to transfer their data from time to time to the respective organisations without any additional drain to exchequer. The data can be analysed in India and can give feed back accordingly. This can also be useful as an alternative means of simple and inexpensive communication between the expeditioners and their families, without any limitation on the number of words per message.

Associated and assisted Army Team to carry out repair and maintenance of gen sets and life supporting equipments.

Recommendations

Some of the salient observations made during the period of stay and suggestions/recommendations are given below:

1. Load distribution system has been studied and redistributed to reduce the imbalance per phase. After redistribution it has been observed that the imbalance is within the permissible limit of 20 to 30 per cent.
2. Control panel of boilers (4 Nos) had no time totalizers. The panels were modified and time totalizers installed separately for recording the running of burner and boiler.
3. Pumping sets installed in the boiler room have been realigned and installed with proper pipe connection to enable easy operation, maintenance and to have more working space.
4. To avoid fire hazards, PVC cables wherever used were replaced with PTFE Teflon insulated cables. Old casing/capping were replaced with new ones to secure the cables properly in main corridor. Complete rewiring has been carried out in pump house and Klargest room.
5. It was observed that the exit from B block was being used frequently. This door also had gaps in between the frame. Due to this, the cold wind rushed inside this Block and toilet area and complete corridor became colder every time the door was opened, and it was difficult to control the temperature. An air lock chamber of size 2.35 m × 1.25 m × 2.5 m was designed and constructed adjacent to the exit door and the temperature could be controlled.
6. The power supply cable from Aditya to main distribution panel of A block, A block to boiler room and summer hut area were properly secured to avoid any damage during high speed wind and blizzards.
7. During the firing of the incinerator toilets, foul odour was escaping through the gaps of the door/frame and was spreading to the complete corridor and nearby area. After replacing door and exhaust fan, this could be controlled.
8. Prefabricated components and panels of light weight are best suited for construction of any habitation in Antarctic conditions for easy handling and construction with available manpower. Panels shall be of standardised size.

9. Structures shall be constructed on stilts or on raised platform to allow free passage of snow drift and to avoid snow accumulation.
10. For assembly of structures, minimum hardware such as nuts, bolts, screws, etc. are to be used to reduce the effort, save time and restrict snow ingress.
11. For inner lining of the containerised accommodation, prefabricated fire retardant panels with proper insulating material are suggested for easy fixing and assembly.
12. Louvres of exhaust fans shall have provision for closing and locking the baffle plates air tight, when not in use to avoid entry of cold air and snow.
13. Even though a fire accident occurred in the old gen set accommodation, it is felt that generators shall be in one power house which will facilitate to carry out easy maintenance, repairs, attendance and periodic observation. This can also eliminate quite a number of cable connections and prevent sparking and accidents. However, an alternative power source may be available to meet the emergency demands.
14. At present 2×62.5 kVA gen sets are running at a time to meet the power requirements. Higher capacity fuel efficient gen sets can be used in place of these, which will reduce the inventory, and can give better output and service. Running of one gen set is always better than two, considering the fact of fuel saving, maintenance, running, etc.
15. New accommodation for the following is considered essential for smooth functioning of Maitri station:
 - (i) There is no proper space available for storing the various spare parts, food articles, hardwares, perishable items, paints, etc. Most of these items are being kept outside. During winter, snow accumulates over these items causing difficulty in finding out and retrieving when the need arises and also get spoiled when these items are kept outside for a long time.
 - (ii) Scientist Lab : Scientists of wintering team and summer team find it extremely difficult to install their equipment/instruments for experiments since no proper space is available. During X wintering, space available in the main entrance was converted to use as NIO lab, which shows the scarcity of space. Also, the need of a drawing board and necessary material is felt essential.
 - (iii) Computer Room : At present the computer is installed in Station Commander's office. Frequent operation of computer and printer by members causes inconvenience and disturbance to leader to

carry out his work, and the work time slot is limited to the normal office hours only, since the adjacent room is occupied by the leader.

- (iv) Guest Room : Lounge is living-cum-dining hall and whenever any guests arrive, there is no place available for sitting and entertain them. Guest room with seating arrangement and all amenities is felt essential.
- (v) Laboratory located at summer camp area does not have any proper communication facility. Difficulty is experienced in contacting the scientists and personnel working in that area. In case of any emergency, no contact with main station is possible, therefore a permanent communication system should be provided.

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